

# Assessing the Robustness of TerraMind to Sensor Domain Gap for On-board Processing on the $\Phi$ -sat-2 Mission

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## Abstract

On-board processing promises to overcome the downlink latency and bandwidth constraints that limit the usefulness of Earth observation (EO) data for time-critical applications such as flood and wildfire response. Geospatial Foundation Models (GeoFMs) are attractive candidates for this role because their large-scale, multimodal pre-training yields semantic feature extractors that generalize across tasks with few labeled examples. However, GeoFMs are pre-trained on archived data from missions such as Sentinel-2, and their behavior when exposed to a new sensor is unknown. This thesis investigates whether TerraMind, a Vision Transformer (ViT)-based GeoFM, produces domain-invariant latent representations when applied to real imagery from the European Space Agency’s  $\Phi$ -sat-2 CubeSat mission, and whether that invariance can be induced through domain adaptation and preserved through compression to a hardware-deployable architecture.

Using a co-registered triplet dataset of real  $\Phi$ -sat-2, real Sentinel-2, and physics-simulated  $\Phi$ -sat-2 imagery, we show that the domain gap between the two sensors is measurable and concentrated in early encoder layers, driven primarily by differing spectral response functions rather than by spatial resolution mismatch. We further show that naively fine-tuning TerraMind on physics-degraded data does not reduce this gap relative to an undegraded baseline, motivating an explicit, paired-contrastive domain-invariance objective that exploits the co-registration directly. This mechanism is applied during knowledge distillation of TerraMind into a lightweight convolutional neural network (CNN) suitable for the  $\Phi$ -sat-2 on-board processor, addressing the incompatibility of ViT architectures with space-qualified hardware. The resulting pipeline is evaluated on real-data land-cover classification and on a physics-degraded flood-segmentation benchmark, and compared against  $\Phi$ satNet, a bespoke on-orbit GeoFM developed for the same mission, to assess whether adapting an existing large-scale pre-trained model offers a label-efficiency advantage over training a mission-specific model from scratch. The results position deliberate, encoder-level domain-invariance training – rather than incidental exposure to degraded data – as a necessary condition for realizing the generalization promise of GeoFMs in resource-constrained, on-board Earth observation settings.

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# Chapter 1

## Introduction

The amount of data that Earth observation (EO) satellites capture exceeds the capacity of ground segments to downlink and process it in a timely manner[1]. Though the data is processed on the ground once it has been fully downlinked, some applications such as flood mapping[2] and wildfire detection[3] require near-real-time inference, because the events they monitor evolve on timescales of hours. Moving the inference step from the ground to a satellite with on-board processing capabilities allows to downlink small data products instead of large raw imagery. This paradigm shift removes the downlink bottleneck for time-critical applications, and a growing number of missions have started demonstrating that it is technically feasible[1]. The European Space Agency’s  $\Phi$ -sat-2 mission is among the most complete demonstrations to date: a 6U CubeSat carrying an 8-band high-resolution multispectral imager alongside the Ubotica CogniSAT-XE1 payload, built around the space-qualified Intel Movidius Myriad 2 Vision Processing Unit (VPU) [4]. Rather than running a single fixed application,  $\Phi$ -sat-2 uses the NanoSat MO Framework[5] to install, update, and orchestrate multiple AI applications on board, much as a smartphone runs apps rather than a single program.

To date, running applications on board of  $\Phi$ -sat-2 have been small, task-specific convolutional neural network (CNN) trained, validated, and uplinked individually for specific jobs such as cloud masking, vessel detection and roadmap transformation[6]. Adding a new capability means repeating the entire pipeline from scratch. Geospatial Foundation Models (GeoFMs) offer a different model of development. Pre-trained on large, diverse archives of EO imagery, a GeoFM such as TerraMind[7] learns a general-purpose semantic representation that downstream tasks can exploit with comparatively few labeled examples, rather than requiring a bespoke architecture and full training run per application. This property matters specifically for applications like flood and burnt-area mapping, where labeled events are rare by nature and a mission cannot simply wait to accumulate a large annotated archive before it becomes useful in an emergency.

Two obstacles stand between this promise and an operational on-board GeoFM. The first challenge is the model architecture, which is related to hardware constraints. GeoFMs built on Vision Transformers (ViTs)[8] are computationally and structurally incompatible with space-qualified accelerators of the Myriad 2’s generation[4], which pre-date the hardware and software support ViT inference requires. The second challenge is the operational domain gap. GeoFMs such as TerraMind are pre-trained on Sentinel-2 and similar archived missions;  $\Phi$ -sat-2 is a physically different sensor, and the statistical properties of its imagery differ from Sentinel-2’s in ways that a model never saw during pre-training. Solving either problem alone does not solve the other: a compressed model

built without addressing the domain gap simply compresses the gap along with it, and a domain-adapted ViT still cannot run on the target hardware.

This thesis addresses both obstacles together, using TerraMind and  $\Phi$ -sat-2 as the concrete case study, with flood segmentation as the motivating downstream application. The methodological approach combines three elements. First, a physics-based sensor degradation pipeline, adapted from the simulator developed for the ESA  $\Phi$ -lab OrbitalAI challenge [9], is used to project labeled Sentinel-2 imagery into the  $\Phi$ -sat-2 sensor domain, producing labeled training data for a target sensor that has not itself accumulated labeled flood events. Second, a paired-contrastive domain-adaptation objective exploits a purpose-built dataset of spatially and temporally co-registered image triplets – real  $\Phi$ -sat-2, real Sentinel-2, and simulated  $\Phi$ -sat-2, all of the same scene – to train the model to represent the same underlying content identically regardless of which sensor produced it. Third, this domain-invariance objective is applied directly during knowledge distillation[10] of the TerraMind encoder into a lightweight CNN, so that the deployable student model inherits sensor-invariance as a property of how it is trained, rather than as an incidental byproduct of compression.

## 1.1 Problem Definition

The main technical problem is a covariate shift between the sensor domain a GeoFM is pre-trained on and the sensor domain it is tasked to operate on. Let  $X$  denote the space of satellite image patches and  $Y$  the space of semantic labels (land-cover class, flood/background). Writing  $p_s$  and  $p_t$  for the source (pre-training, e.g. Sentinel-2) and target (operational, e.g.  $\Phi$ -sat-2) distributions, a covariate shift is present when

$$p_t(x) \neq p_s(x) \quad \text{while} \quad p_t(y | x) = p_s(y | x). \quad (1.1)$$

That is, the physical scene and its semantic content are identical across the two domains – the same flooded field is still a flooded field – but the pixel statistics produced by the sensor observing it differ, because of differing spectral response functions, point spread functions, signal-to-noise characteristics, and geometric band-to-band alignment. A model whose learned representation is sensitive to these sensor-specific statistics rather than to the underlying semantics will degrade on the target domain even though nothing about the task itself has changed.

Prior work on GeoFM domain gap has typically measured this shift only indirectly, through downstream task performance, and has compared source and target datasets that differ in geography, acquisition date, and scene content in addition to sensor – confounding the sensor-specific shift this thesis is concerned with, with shifts due to the underlying data distribution itself. The co-registered triplet dataset used in this work is designed specifically to remove that confound, since the real  $\Phi$ -sat-2, real Sentinel-2, and simulated patches used for evaluation depict the same scene at the same time.

## 1.2 Research Gap

Two bodies of closely related work sharpen, rather than close, the gap this thesis addresses. Du et al. [11] recently demonstrated an on-orbit GeoFM (a compressed Prithvi variant) on the Kanyini CubeSat and the IMAGIN-e payload, adapting it to the target sensor by retraining task heads on real, though limited, target-domain labels while

keeping the encoder frozen. Their measured domain gap, however, is confounded in the way described above: source and target are entirely different benchmark datasets, not co-registered pairs, so sensor-induced shift cannot be isolated from content and geographic shift. Separately,  $\Phi$ satNet, a bespoke U-Net-based GeoFM developed within the same research group specifically for on-board execution on  $\Phi$ -sat-2 hardware, sidesteps the ViT-compatibility problem altogether by training a compact, non-transformer architecture from scratch on a purpose-built pre-training corpus, rather than adapting an existing large-scale pre-trained model. Neither work isolates sensor-only covariate shift using co-registered data, and neither tests whether an existing GeoFM’s broad, diverse pre-training transfers to a new sensor with fewer labels than a mission-specific model trained from scratch requires – the specific comparison this thesis undertakes against  $\Phi$ satNet in Chapter 4.

The gap this thesis addresses is therefore threefold: (i) whether the domain gap between  $\Phi$ -sat-2 and Sentinel-2-pretrained TerraMind can be measured directly, in latent space, using data that isolates sensor-induced covariate shift from confounding factors; (ii) whether that gap can be closed through an explicit, paired-contrastive domain-invariance objective rather than incidental exposure to degraded training data; and (iii) whether the resulting invariance survives compression into a hardware-deployable, cross-architecture (ViT-to-CNN) student model.

### 1.3 Research Objective and Problem Statement

The objective of this thesis is to determine whether TerraMind can be made to produce domain-invariant latent representations under the  $\Phi$ -sat-2 sensor domain gap, and whether this invariance can be preserved through knowledge distillation into a lightweight CNN suitable for deployment on the  $\Phi$ -sat-2 on-board processor, without sacrificing the representational benefits of large-scale GeoFM pre-training. This objective is pursued through three research questions:

1. Does the domain gap caused by sensor differences between  $\Phi$ -sat-2 and Sentinel-2 measurably affect the latent representations produced by TerraMind, and if so, where in the encoder does this effect concentrate?
2. Can TerraMind be trained to produce more domain-invariant representations using a domain-adaptation technique that exploits co-registered, physics-simulated sensor pairs, given that naive fine-tuning on degraded data alone does not achieve this?
3. Can the resulting domain-invariant encoder be distilled into a lightweight CNN suitable for on-board  $\Phi$ -sat-2 inference while preserving the learned invariance, rather than only preserving task accuracy?

Answering these questions would show that the generalization promise of large GeoFMs – broad semantic transfer from limited labels – is compatible with the low-latency, resource-constrained inference that on-board emergency-response applications require, provided the domain gap is addressed deliberately rather than assumed away.

# Chapter 2

## Background and Literature Review

This chapter positions the thesis against five parts of the work that each contribute to the overall problem, which is to deploy a robust GeoFM on board a resource-constrained satellite. Each part is treated in turn, before a closing synthesis states what combination of them remains unaddressed.

### 2.1 Geospatial Foundation Models

GeoFMs extend the foundation-model paradigm from natural images to Earth observation, pre-training a single large backbone on diverse, multi-modal EO archives so that many downstream tasks can be solved by fine-tuning a small task-specific head rather than training a full model from scratch. TerraMind[7] is representative of this generation: a ViT-based, multimodal GeoFM pre-trained across nine geospatial modalities at scale, explicitly designed to produce transferable semantic representations rather than task-specific features. Studies compressing and deploying such models, discussed further below, treat this generalization capacity as the primary reason to prefer a GeoFM over a bespoke architecture – but that capacity has, to date, mostly been demonstrated on data drawn from the same sensor family the model was pre-trained on, leaving open how it behaves under a genuine sensor change.

### 2.2 Domain Adaptation and the Sensor Domain Gap

Domain adaptation addresses the general problem of a model trained on a source distribution being applied to a target distribution with different input statistics but unchanged semantics [12]. Within satellite on-board AI specifically, Du et al. [13] demonstrated domain adaptation for satellite-borne cloud detection by retraining task heads on real, though limited, labeled imagery from the Kanyini mission, with the encoder frozen. This line of work was extended on-orbit by Du et al. [11], who deployed a compressed Prithvi-family GeoFM on Kanyini and the IMAGIN-e payload, again adapting via real-label head retraining rather than encoder-level invariance training, and measuring the domain gap only indirectly through downstream task performance on source and target benchmark datasets that differ in geography and content as well as sensor. This is the central limitation this thesis’s data design addresses: without co-registered source/target pairs of the same scene, a measured performance drop cannot distinguish sensor-induced covariate shift from confounding differences in what the two datasets actually depict.

## 2.3 Physics-Based Sensor Degradation Simulation

Because labeled data from a satellite sensor does not exist before the satellite is launched, and accumulates slowly afterward for rare events, physics-based simulation of a target sensor’s characteristics from an existing labeled source (typically Sentinel-2) has become a standard tool for pre-mission model development. The simulator developed for the ESA  $\Phi$ -lab OrbitalAI challenge [9] models this transformation as a sequence of physical operations in radiance space – spectral and spatial resampling to the target sensor’s band configuration and ground sampling distance, band-to-band misalignment consistent with push-broom acquisition geometry, signal-to-noise degradation, and optical blur modeled through the sensor’s modulation transfer function – calibrated against real flight data from related missions. This simulator underlies both the degradation pipeline used in this thesis and, independently, the pre-training and benchmarking datasets used by  $\Phi$ satNet (Section 2.5.1), making it a shared, validated tool within this specific research context rather than a novel contribution of either work.

## 2.4 Knowledge Distillation Across Architectures

Knowledge distillation [10] compresses a large teacher model into a smaller student by training the student to reproduce the teacher’s output distribution, or, in feature-based variants, its intermediate representations, rather than training the student only against hard labels. Most applications of distillation to on-board EO models compress within the same architecture family or distill into a same-family, smaller model, and treat robustness to sensor conditions as a property that transfers passively if the teacher itself is robust. This thesis instead distills across architecture families – from a ViT-based teacher to a CNN student – a necessary step given that ViT inference is not supported on the  $\Phi$ -sat-2 on-board hardware (Section 2.5), and applies the domain-invariance objective directly to the student during distillation rather than relying on passive inheritance from an already-invariant teacher.

## 2.5 On-board AI for Earth Observation

On-board AI processing has moved from single-purpose CNNs, exemplified by  $\Phi$ -sat-1’s cloud-detection application[14], toward more general and more capable models, as surveyed by Barretta et al. [1]. The hardware reality constraining this progress is specific and severe: the  $\Phi$ -sat-2 CogniSAT-XE1 payload’s Intel Movidius Myriad 2 VPU [4] is legacy hardware whose supported operator set (through OpenVINO 2020.3, ONNX opset 8–11) lacks dynamic-axes support, making ViT inference structurally infeasible regardless of model size, and imposes hard memory (approximately 512 MB) and model-size (250 MB) limits under float16-only inference. These constraints, documented in detail by the same-lab  $\Phi$ satNet effort, are the concrete justification for why this thesis treats cross-architecture distillation, rather than direct deployment or same-family compression, as a requirement rather than a design choice.

### 2.5.1 $\Phi$ satNet: a different approach

$\Phi$ satNet is a U-Net-derived, multi-task-pretrained GeoFM developed within the same research group specifically for on-board execution on  $\Phi$ -sat-2, using a purpose-built pre-training corpus and benchmark suite built on the same OrbitalAI simulator described above. By training a compact, non-transformer architecture from scratch,  $\Phi$ satNet avoids the ViT-hardware-incompatibility problem entirely, and its authors report that it matches or exceeds unconstrained transformer-based GeoFMs on several on-board tasks. This is a different design philosophy from the one pursued here, not a competing solution to the same problem:  $\Phi$ satNet optimizes for a hardware-native architecture trained on a curated, mission-specific task suite, while this thesis bets that adapting an existing, diversely pre-trained GeoFM generalizes to new tasks with fewer labels than training a mission-specific model affords. Chapter 4 tests this bet directly through a few-shot comparison against  $\Phi$ satNet.

## 2.6 Synthesis

None of the five bodies of work above, individually or in combination, isolates sensor-induced covariate shift using co-registered source/target data, trains encoder-level domain invariance through an explicit paired objective rather than incidental exposure to degraded or limited real data, and verifies that this invariance – not merely task accuracy – survives cross-architecture distillation into hardware-deployable form. The spatially and temporally co-registered  $\Phi$ -sat-2/Sentinel-2/simulated triplet dataset used in this thesis is what makes the first of these three possible; the paired-contrastive distillation objective developed in Chapter 3 is what makes the second and third possible.

# Chapter 3

## Methodology

This chapter presents the complete analytical workflow: the data used and the role each dataset plays, the physics-based degradation pipeline and its calibration, the protocol for measuring the domain gap, the domain-adaptation mechanism adopted (and the one rejected, with the evidence for rejecting it), the cross-architecture distillation procedure, and the evaluation protocol. It closes with a dedicated account of the tooling and reproducibility practices used throughout, as required by the thesis guidelines.

The data underlying this work is split into two categories. The first is the co-registered triplet dataset itself: real  $\Phi$ -sat-2 acquisitions, obtained via the ESA Insula Perception platform, matched to concurrent Sentinel-2 scenes and to a physics-simulated  $\Phi$ -sat-2 rendering of the same Sentinel-2 scene, co-registered using the LightGlue deep feature-matching algorithm [15] and filtered for spatio-temporal proximity and cloud cover. This dataset carries land-cover labels derived from ESA WorldCover 2021, providing the only currently available real-sensor ground truth. The second category is simulated: a physics-degraded, graded version of the Sen1Floods flood-segmentation benchmark, produced with the same degradation pipeline, standing in for a real labeled  $\Phi$ -sat-2 flood dataset that does not yet exist because the mission has not captured an annotated flood event.

The overall workflow followed a prototyping-implementation-refinement logic rather than a fixed plan executed once. An initial conceptual model – fine-tune TerraMind on real  $\Phi$ -sat-2 labels, measure the gap, close it – was implemented, evaluated against intermediate diagnostics (embedding visualizations, similarity metrics), and revised twice on the basis of what those diagnostics showed: first when a normalization defect was found to be corrupting an early performance baseline (Section 3.3.1), and second when a first domain-adaptation mechanism was found, empirically, not to work (Section 3.4.1), motivating the mechanism actually adopted (Section 3.4.2).

### 3.1 Data

#### 3.1.1 The co-registered triplet dataset

The main dataset is a collection of spatially and temporally co-registered image triplets, produced in collaboration with the ESA  $\Phi$ -lab: real  $\Phi$ -sat-2 acquisitions (obtained via the ESA Insula Perception platform), the corresponding real Sentinel-2B scene, and a physics-simulated  $\Phi$ -sat-2 rendering of that same Sentinel-2 scene (Section 3.2). Candidate scenes were filtered by spatio-temporal proximity between the  $\Phi$ -sat-2 and Sentinel-2 overpasses and by a cloud-cover threshold on both acquisitions, and precise pixel-level alignment

was achieved using the LightGlue deep feature-matching algorithm [15], which identifies corresponding semantic keypoints across the two sensors despite their differing spatial and radiometric characteristics.

This dataset carries labels for a single task: land-use/land-cover (LULC) classification, derived from ESA WorldCover 2021 and rasterized onto each triplet’s real- $\Phi$ -sat-2 footprint. Its role in this thesis is twofold: it is the only real-sensor ground truth available for any downstream task, and its co-registration is what makes it possible to measure sensor-induced covariate shift in TerraMind’s latent space (Section 3.3) without the scene-content and geography confounds present in prior domain-gap studies that compare unrelated source and target datasets.

Figure 3.1 shows four such triplets. Qualitatively, the real acquisitions are consistently grainier than their simulated or source counterparts, and occasionally differ subtly in overall tone; the simulated column is, by construction, closer in appearance to Sentinel-2 than to the real sensor it is meant to approximate, a point returned to below.

### Scale and coverage

After discarding a small number of corrupted source products, the dataset comprises 253,228 co-registered triplet patches drawn from 1,299 distinct  $\Phi$ -sat-2 products (individual acquisitions), a median of 209 patches per product (up to 713 for the largest). Acquisitions span 2024-08-21 to 2026-04-15 (21 months), with collection volume markedly higher from mid-2025 onward than in the initial ramp-up period. Real- $\Phi$ -sat-2 and Sentinel-2 acquisition dates are matched to within a median of 6.8 days and a maximum of 14.9 days, not simultaneously, which is an acceptable temporal offset for a largely stable target such as land cover but a caveat worth stating rather than assuming away. 73.1% of retained patches have under 5% total cloud cover, and only 0.199% of pixels are unlabelled after the ESA WorldCover rasterization step.

Geographically, the 1,299 source products span latitudes  $-54^\circ$  to  $76^\circ$  with global but uneven coverage, concentrated over western North America, the southern Andes, the Mediterranean basin and Middle East, and parts of East and South Asia (Figure 3.4). By Köppen–Geiger climate group, the patch population is dominated by arid (41%) and temperate (29%) climates, with tropical (14%), continental (10%), and polar (1%) regions comparatively under-represented (4% unclassified).

The eleven ESA WorldCover classes present are similarly imbalanced by construction rather than by sampling choice: grassland (20.8%), tree cover (20.0%), bare/sparse vegetation (16.2%), built-up (14.4%), and cropland (13.4%) account for the large majority of labelled pixels, while snow/ice, mangroves, and moss/lichen each fall below 0.6% (Figure 3.5). The typical patch is fairly homogeneous – a single class occupies a median of 73% of any given patch’s labelled area.

### Train/validation/test split and a limitation

The dataset is split 80/10/10 (202,582 / 25,323 / 25,323 patches) at the *patch* level with a fixed seed, and the resulting class-prior distribution is stable across all three splits, confirming the split does not introduce label-distribution shift (Figure 3.6, left). It does, however, introduce a different problem: because splitting is patch-level rather than product-level, all 1,299 source products are represented in the training set, and every one of the 1,287 distinct products contributing to the test set also contributes patches to training (Figure 3.6, right). Since patches from the same product are spatially adjacent

Co-registered views, independently contrast-stretched

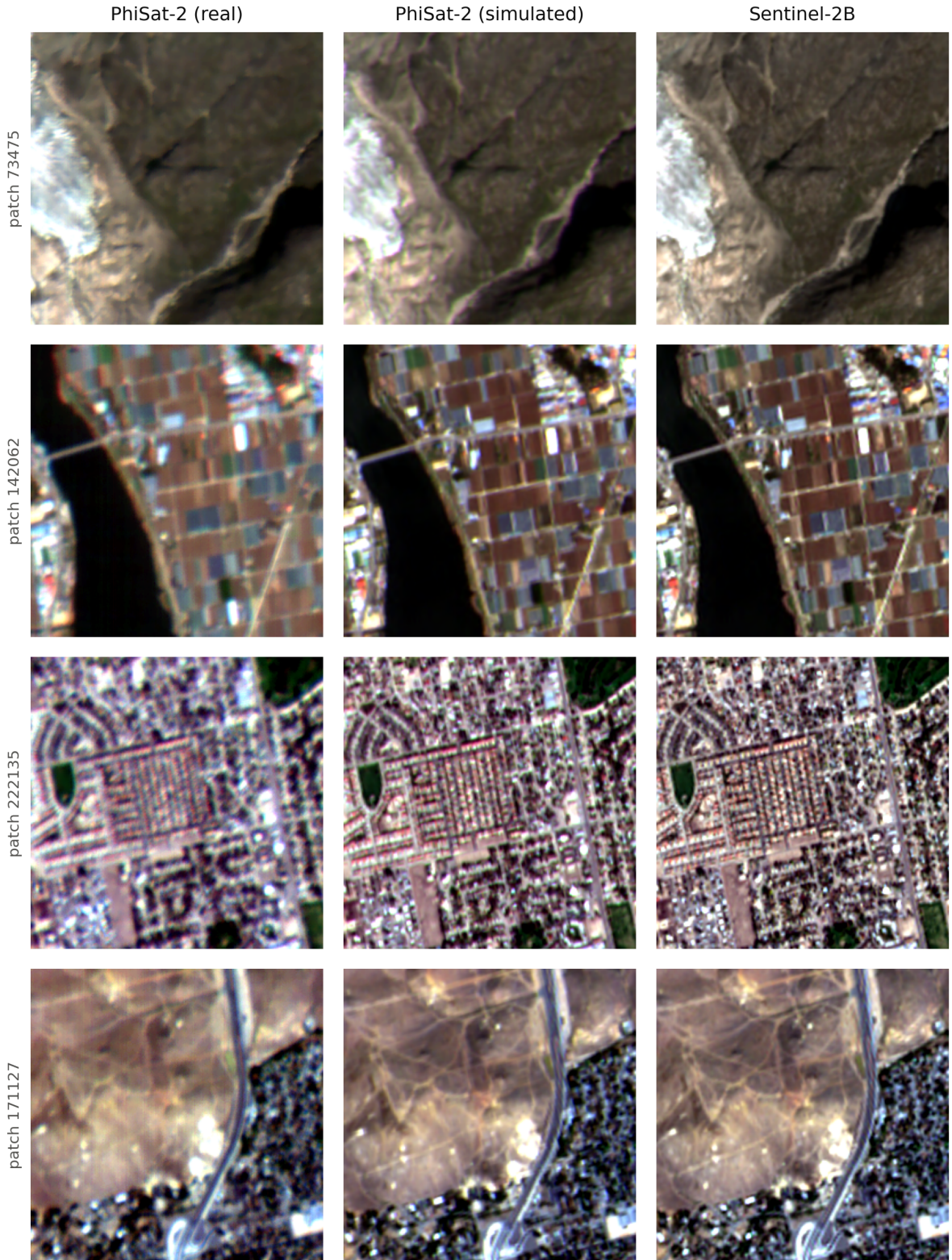


Figure 3.1: Four representative triplets, each row showing the same scene as observed by real  $\Phi$ -sat-2, the physics-simulated  $\Phi$ -sat-2 rendering, and Sentinel-2B (each channel independently contrast-stretched for display). The real acquisitions show visibly grainier, noisier texture than either the simulated or source Sentinel-2 view – most apparent in the dark water/shadow region of patch 171127 – consistent with the sensor noise characteristics the degradation pipeline (Section 3.2) is designed to reproduce.

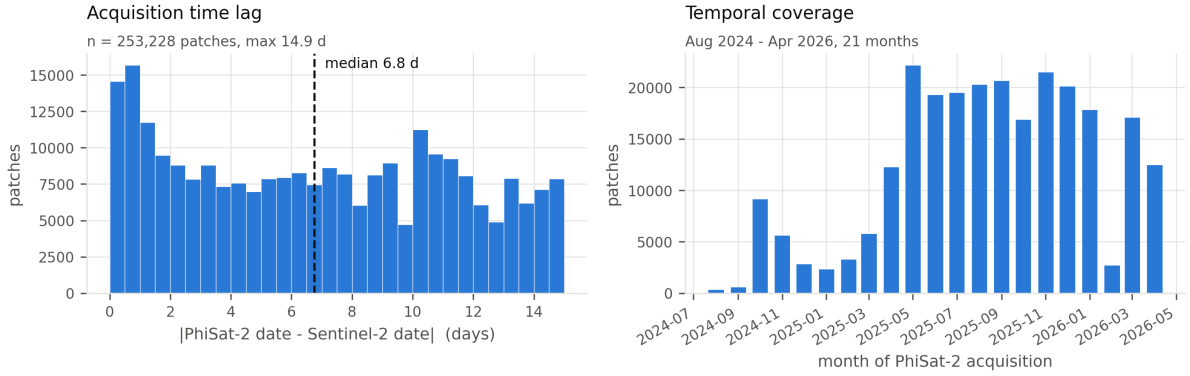


Figure 3.2: Left: distribution of the absolute time lag between matched  $\Phi$ -sat-2 and Sentinel-2 acquisitions (median 6.8 days, maximum 14.9 days,  $n = 253,228$ ). Right: monthly acquisition volume, Aug. 2024 – Apr. 2026, showing the collection ramp-up in the first eight months.

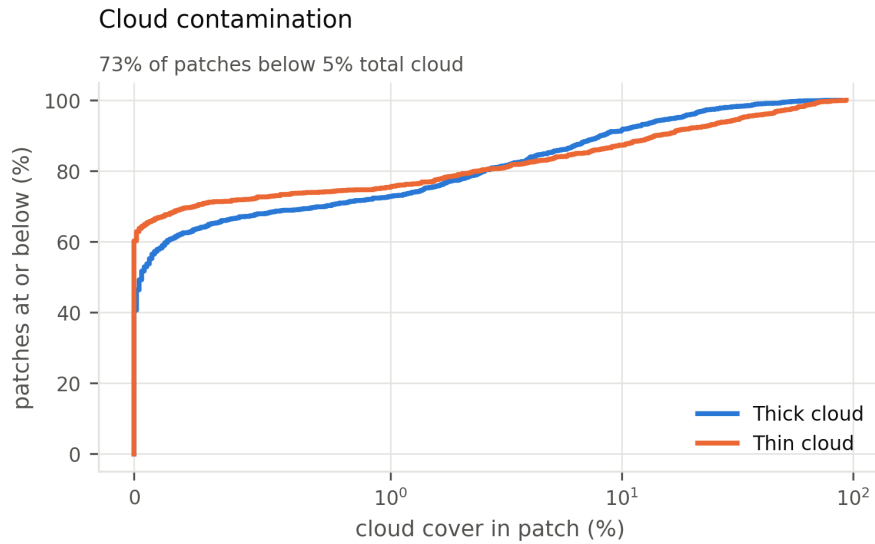


Figure 3.3: Cumulative distribution of per-patch cloud cover, separating thick and thin cloud; 73% of patches fall below 5% total cloud cover.

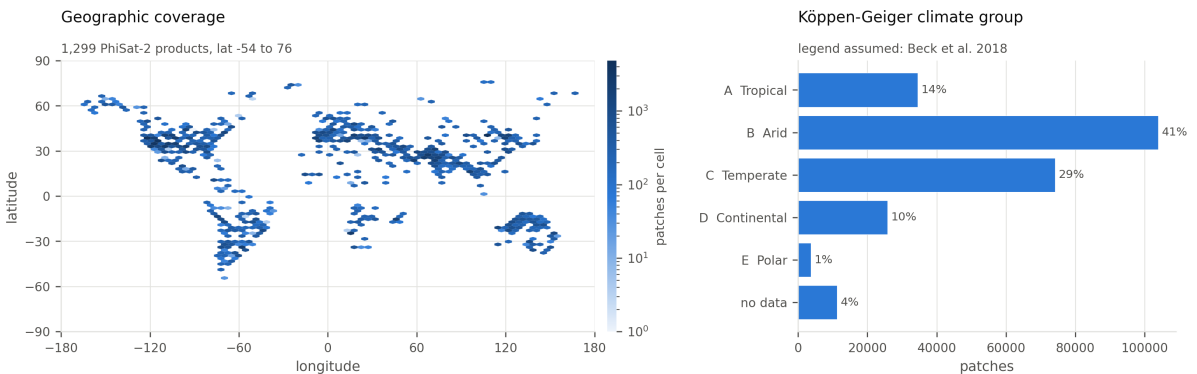


Figure 3.4: Left: spatial density of  $\Phi$ -sat-2 products (hexbin, log-scaled patch count per cell). Right: distribution of patches by Köppen–Geiger climate group (Beck et al., 2018 classification).

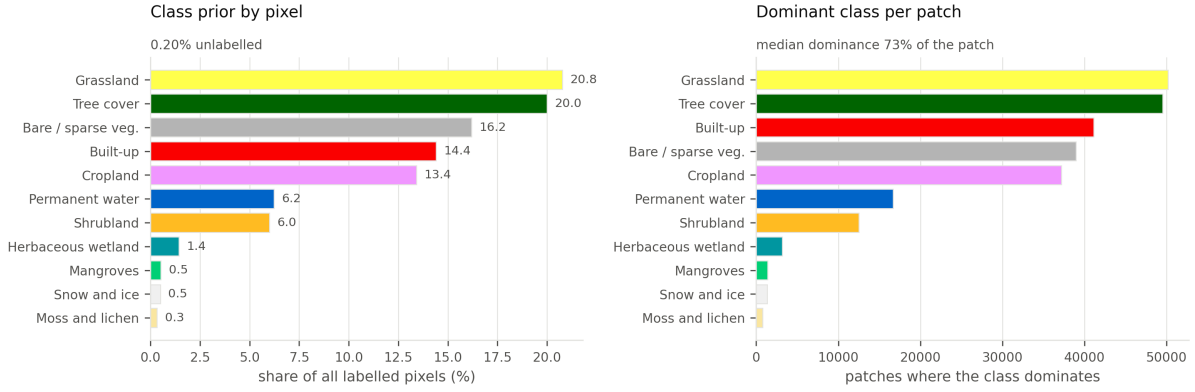


Figure 3.5: Left: share of all labelled pixels by ESA WorldCover class. Right: number of patches in which each class is the dominant (plurality) class.

or overlapping crops of one acquisition, this means the test set is not a clean geographic or scene-level hold-out, and real-data performance figures reported in Chapter 4 should be read as an optimistic upper bound on generalization to genuinely unseen scenes rather than as a strict out-of-sample estimate. A product-level (rather than patch-level) re-split is the direct remedy and is noted as future work in Section 4.4.

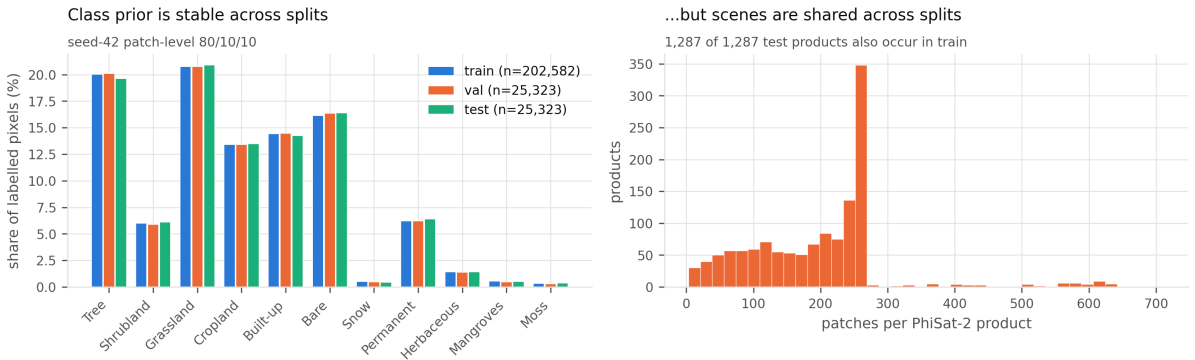


Figure 3.6: Left: class prior is stable across the train/validation/test split. Right: every one of the 1,287 products contributing to the test set also contributes patches to the training set – the split is patch-level, not scene-level.

## Measured sensor discrepancy

Three independent measurements quantify the sensor discrepancy this dataset is designed to expose, prior to any model being involved.

First, the raw radiometric scales of the two sensors differ by roughly an order of magnitude (Figure 3.7): real  $\Phi$ -sat-2 digital numbers have per-band medians of roughly 55–150 (12-bit range, with measurable saturation of 2.1–2.2% of pixels in the RE1/RE2 bands and dropout of 0.3–0.36% in RE2/RE3), against Sentinel-2 and simulated- $\Phi$ -sat-2 medians around 2,200–3,300 (consistent with the L1C TOA-reflectance-scaled DN convention). The simulated and raw Sentinel-2 domains are close to statistically indistinguishable at this level (their boxes are visually superimposed in Figure 3.7), confirming that the physics-based degradation pipeline (Section 3.2) perturbs fine spatial and spectral struc-

ture rather than gross intensity statistics – the domain gap this thesis is concerned with is a real-vs-simulated gap, not a simulated-vs-source gap.

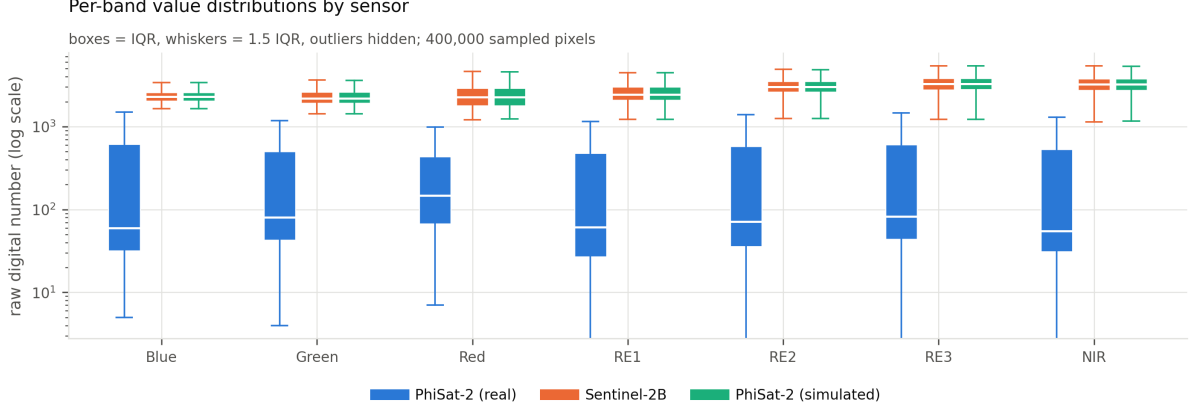


Figure 3.7: Per-band raw digital number distributions by sensor (log scale; boxes = IQR, whiskers =  $1.5 \times \text{IQR}$ , outliers hidden; 400,000 sampled pixels). Real  $\Phi$ -sat-2 occupies a scale roughly one order of magnitude below Sentinel-2 and its simulated counterpart, which closely overlap each other.

Per-domain normalization (Section 3.3.1) closes most, but not all, of this gap (Figure 3.8): after  $z$ -scoring each sensor independently, the three distributions are centered similarly, but real  $\Phi$ -sat-2 retains visibly wider spread and longer tails in every band than Sentinel-2 or the simulated domain. This residual spread difference, not the location/scale difference normalization already removes, is what the domain-adaptation mechanism of Section 3.4 must close. The statistics underlying this figure were measured on the sampled patches shown here, which differ slightly from the constants currently hardcoded in the dataset loader; the loader’s normalization constants should be refreshed from the full measured statistics before the next training run, so that all reported results use the same normalization the figures were generated from.

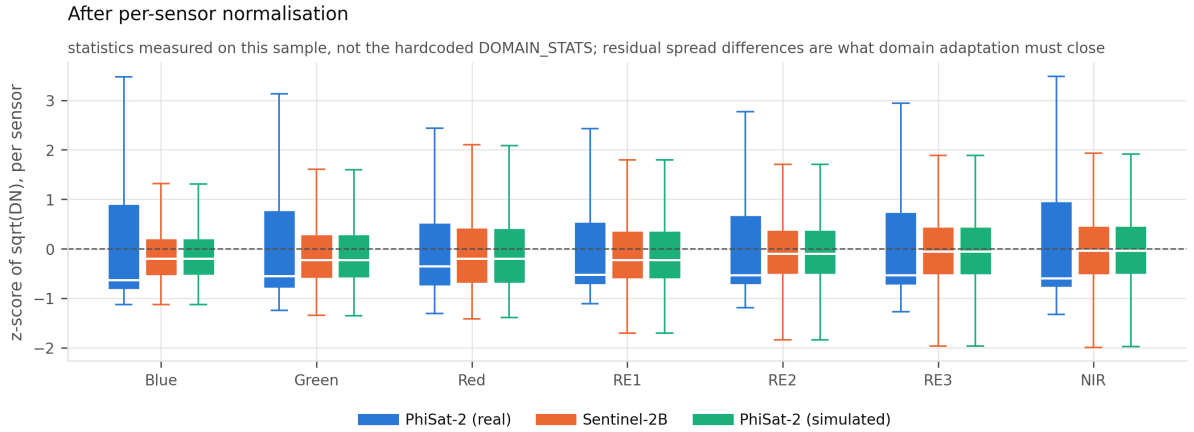


Figure 3.8: The same per-band distributions after per-sensor  $z$ -score normalization of  $\sqrt{\text{DN}}$ . Location and scale are aligned, but real  $\Phi$ -sat-2 retains a wider spread than Sentinel-2 or the simulated domain in every band – the residual gap normalization alone cannot close.

Second, per-band correlation between co-registered real- $\Phi$ -sat-2 and Sentinel-2 pixels is only weak-to-moderate even for matched same-scene observations (Pearson  $r$  0.31–0.54, Spearman  $\rho$  0.37–0.61 across the seven shared bands; Figure 3.9), with the weakest correspondence in the red-edge bands (RE1, RE2) and the strongest in the red band. This is a direct, model-free demonstration that the two sensors disagree substantially about the same physical target observed within the same fortnight. For contrast, the equivalent cross-band correlation between the simulated domain and its Sentinel-2 source is near-perfect (Spearman  $\rho$  0.98–1.00 on matching bands; Figure ??), as expected since the simulated patches are deterministic transformations of the same Sentinel-2 pixels – the gap between real and simulated  $\Phi$ -sat-2 is therefore not explained by any weakness in the correlation analysis itself.

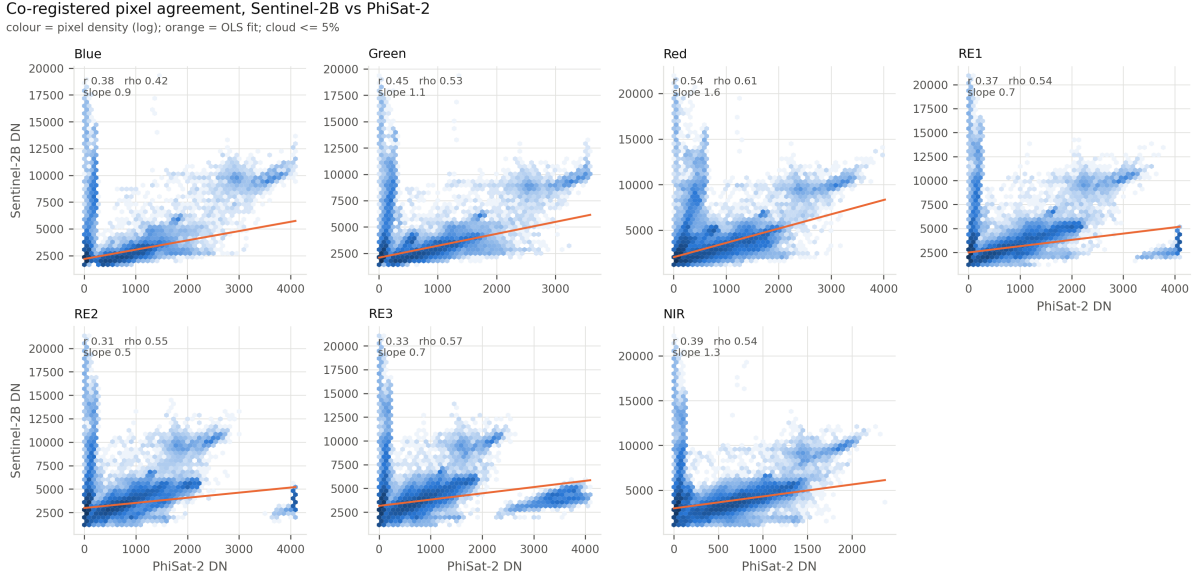


Figure 3.9: Per-band pixel agreement between co-registered real- $\Phi$ -sat-2 and Sentinel-2B digital numbers (hexbin density, log scale; orange = ordinary least-squares fit; patches with  $\leq 5\%$  cloud only). Point clouds are diffuse around the fitted line in every band, and several bands show a distinct vertical streak near zero  $\Phi$ -sat-2 DN spanning the full Sentinel-2 DN range, consistent with localized real-sensor dropout or saturation artifacts rather than measurement noise alone.

Third, and most directly relevant to the degradation-driver hypothesis discussed in Chapter 4, the *shape* of the spectral signature – each sensor’s per-band median normalized by its own overall band mean, so that absolute scale is divided out – differs markedly between real  $\Phi$ -sat-2 and Sentinel-2/simulated- $\Phi$ -sat-2 (Figure 3.10). Sentinel-2 and the simulated domain are, again, visually indistinguishable from each other. Real  $\Phi$ -sat-2, in contrast, shows a pronounced peak at the red band (1.85 relative to its own mean, against 0.85 for Sentinel-2/simulated) and a dip at the near-infrared band (0.70 against 1.21), a difference in relative spectral shape, not merely absolute intensity. This is direct, quantitative support for attributing the domain gap primarily to spectral response function mismatch (Section 4.2) rather than to spatial resolution or noise level, since a resolution- or noise-driven gap would not be expected to alter the relative shape of the spectral signature this distinctly.

These measured per-domain statistics (mean, standard deviation, and clip value per band, computed separately for the real, Sentinel-2, and simulated domains) are what

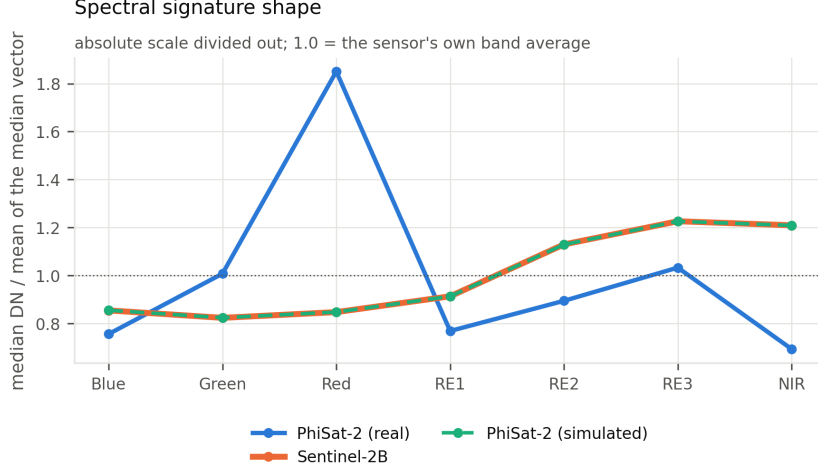


Figure 3.10: Median per-band digital number, normalized by each sensor’s own overall band mean (1.0 = that sensor’s average). Sentinel-2B and simulated  $\Phi$ -sat-2 overlap almost exactly; real  $\Phi$ -sat-2 has a markedly different spectral shape, peaking at Red and dipping at RE1 and NIR.

parametrize the per-domain normalization protocol of Section 3.3.1.

### 3.1.2 Simulated degraded flood-segmentation dataset

No real  $\Phi$ -sat-2 flood dataset exists, because the mission has not yet captured and labeled a flood event; this is a structural limitation of any new mission, not a data-collection oversight. To obtain labeled data for the flood-segmentation task motivating this thesis, the Sen1Floods benchmark is passed through the same degradation pipeline used to build the triplets, producing a graded ladder of  $\Phi$ -sat-2-like flood scenes at increasing degradation severity. This simulated dataset plays a different role from the triplets: it is the training and evaluation substrate for the flood task itself, not the substrate for measuring the domain gap, since it lacks a real- $\Phi$ -sat-2 counterpart against which to validate the simulation directly.

## 3.2 Physics-Based Sensor Degradation Pipeline

The degradation pipeline adapts the simulator developed for the ESA  $\Phi$ -lab OrbitalAI challenge [9], which projects labeled Sentinel-2 L1C tiles into the  $\Phi$ -sat-2 sensor domain by modeling the physical differences between the two instruments as a sequence of operations in radiance space. Because these effects are defined physically on radiance, not reflectance, the pipeline converts source reflectance to top-of-atmosphere radiance, applies each degradation in turn, and converts back to reflectance to yield a simulated L1C product. Pixel coordinates are written  $(i, j)$  and spectral band index  $b_n$  throughout.

**Radiance–reflectance conversion.** TOA reflectance  $\rho$  and TOA radiance  $L$  are related by

$$\rho(i, j, b_n) = \frac{\pi d^2}{E_{\text{SUN}}(b_n) \cos \theta_S(i, j)} L(i, j, b_n), \quad (3.1)$$

where  $d$  is the Earth–Sun distance in astronomical units,  $E_{\text{SUN}}(b_n)$  the exo-atmospheric solar irradiance for band  $b_n$ , and  $\theta_S(i, j)$  the solar zenith angle at pixel  $(i, j)$ . The pipeline enters via the inverse of Equation (3.1) and exits by re-applying it after all radiance-domain degradations; solar irradiance and zenith angle are read from Sentinel-2 tile metadata via the Copernicus Data Space Ecosystem OData API, and no additional solar-irradiance adjustment is needed because Sentinel-2 and  $\Phi$ -sat-2 spectral bands are close enough in wavelength. Earth–Sun distance is computed from acquisition day-of-year as  $d = 1 - 0.01672 \cos(0.9856(\text{day} - 4) \cdot \pi/180)$ .

**Spatial resampling.**  $\Phi$ -sat-2 has a uniform ground sampling distance of 4.75 m across all bands; Sentinel-2 provides 10 m (bands B2, B3, B4, B8) and 20 m (bands B5, B6, B7). Each band is resampled to the common 4.75 m grid using bicubic interpolation, which limits the resampling artifacts that bilinear interpolation would introduce, though it cannot recover spatial detail the coarser source bands never captured.

**Band-to-band misalignment.** Push-broom acquisition co-registers spectral bands imperfectly, since detector rows are exposed at slightly different times and platform micro-vibration perturbs the line of sight during each band’s integration window. For the raw L1A product, the displacement of band  $b_n$  relative to the previously acquired band follows

$$\|\Delta \mathbf{r}(b_n)\| \sim \mathcal{N}(\mu_{b_n}, \sigma^2), \quad \mu_{b_n} \in [0.44, 1.1] \text{ pixels}, \quad (3.2)$$

with direction sampled uniformly over  $[0, 2\pi)$  and displacement accumulating across preceding bands. For the geometrically corrected L1C product used throughout this thesis, an on-board co-registration step removes this accumulation, and the residual is instead modeled as a single random displacement of each band relative to a reference band ( $\mu_{b_n} = 0$ ), with standard deviation set higher over water ( $\sigma_{\text{sea}} = 6$  pixels) than over land ( $\sigma_{\text{land}} = 1$  pixel), reflecting the harder auto-registration problem posed by feature-poor sea surfaces.

**Signal-to-noise degradation.** The baseline formulation draws a per-band noise sample from  $\mathcal{N}(0, 1)$  scaled by the ratio of a reference radiance to a specified SNR,

$$L_{\text{noise}}(i, j, b_n) = \frac{L_{\text{ref}}(b_n)}{\text{SNR}(b_n)} \mathcal{N}(0, 1), \quad (3.3)$$

which implicitly treats noise variance as signal-independent, matching the specified SNR only near  $L_{\text{ref}}$ . A signal-dependent variant, consistent with a shot-noise regime in which SNR scales with the square root of collected signal, rescales the reference SNR to an effective, radiance-dependent value and samples

$$L_{\text{noise}}(i, j, b_n) = \frac{\sqrt{L_{\text{in}}(i, j, b_n) L_{\text{ref}}(b_n)}}{\text{SNR}(b_n)} \mathcal{N}(0, 1). \quad (3.4)$$

The implementation used for the calibrated baseline degradation level applies Equation (3.4). For the extended sweep used to push beyond nominal  $\Phi$ -sat-2 noise conditions – needed to stress-test domain-invariance under more severe degradation than the flight-validated parameters cover – a simplified variant samples a base SNR uniformly from a specified range and scales noise by the ratio of that SNR to the batch-wise maximum radiance, trading exact photometric fidelity for the ability to sweep degradation

severity beyond the calibrated regime. This trade-off is stated explicitly here because it is a deliberate design choice, not an approximation error: the baseline level is flight-parameter-calibrated, and only the extended sweep sacrifices fidelity for range.

**Optical blur via MTF/PSF.** Optical blur is modeled through the system modulation transfer function (MTF), aggregating static (optics, detector), dynamic (jitter, platform instability), and smearing (motion) contributions. Assuming shift-invariant blur, imaging reduces to convolution with the system point spread function (PSF). Given the MTF amplitude at Nyquist frequency  $M(b_n)$ , the MTF is modeled as Gaussian in normalized spatial frequency  $k$ ,

$$\text{MTF}(b_n, k) = e^{\ln M(b_n) \cdot k^2}, \quad (3.5)$$

with the corresponding PSF obtained as its inverse Fourier transform and convolved with the radiance image. For  $\Phi$ -sat-2 the average Nyquist MTF is  $M_\Phi(b_n) \approx 0.05$ ; for Sentinel-2,  $M_{S-2}(b_n) \approx 0.30$ . The reference simulation applies Equation (3.5) directly with  $M_\Phi(b_n)$ , leaving the intrinsic Sentinel-2 blur uncorrected; a combined-filter form exists for the case where source blur should also be compensated, but is not required here since the intrinsic Sentinel-2 blur is small relative to  $\Phi$ -sat-2's.

This pipeline is shared infrastructure within the research group: it also underlies the pre-training and benchmark datasets used by  $\Phi$ satNet (Section 2.5.1). The novelty of this thesis lies in how the pipeline's output is used – to isolate and then close sensor-specific covariate shift via co-registered pairs – not in the pipeline itself.

## 3.3 Domain Gap Measurement

### 3.3.1 Normalization protocol

Raw  $\Phi$ -sat-2 digital numbers are heavy-tailed; a square-root transform is applied first to bring the distribution closer to Gaussian, followed by a smooth (soft-saturating) clip toward a domain-specific clip value rather than a hard clip, and finally per-domain  $z$ -score standardization using mean and standard deviation statistics computed separately for each sensor domain (real  $\Phi$ -sat-2, real Sentinel-2, simulated  $\Phi$ -sat-2), rather than a single shared normalization applied across domains. An early baseline was corrupted by a normalization defect that caused weight saturation and an artificially depressed performance estimate; correcting it was a precondition for every result reported in Chapter 4.

Because per-domain normalization could, in principle, itself account for an apparent domain gap that is really just a difference in raw value scale, the domain-gap measurement below was run both with and without per-domain normalization. The two conditions produced similar overall results, and per-domain normalization in fact produced *lower* cosine similarity between domains in the encoder's early layers – the opposite of what a normalization artifact would predict – indicating that the measured gap in early layers is a genuine representational effect rather than a calibration artifact.

### 3.3.2 Direct and indirect evidence

The domain gap is measured directly, in latent space, using the co-registered triplets: Maximum Mean Discrepancy (MMD) between the embedding distributions of matched real- $\Phi$ -sat-2, real-Sentinel-2, and simulated- $\Phi$ -sat-2 patches, and t-SNE visualization of

the same embeddings, both computed per encoder layer to locate where the gap concentrates. It is measured indirectly through downstream task performance: LULC mIoU on real  $\Phi$ -sat-2 data relative to a Sentinel-2-only fine-tuned baseline evaluated zero-shot on the same real data, and flood-segmentation IoU across the simulated degradation ladder of Section 3.1.2.

## 3.4 Domain-Adaptation Mechanism

### 3.4.1 A rejected mechanism: naive exposure to degraded data

The first mechanism attempted was implicit: fine-tune TerraMind’s encoder on the LULC task using degraded, simulated training data, on the premise that exposure to degraded examples during training would itself induce a more domain-invariant representation. Cross-validation against the real- $\Phi$ -sat-2/Sentinel-2 triplets showed this does not hold: a model fine-tuned this way shows no lower domain gap to real  $\Phi$ -sat-2 than a model fine-tuned only on clean Sentinel-2. Standard supervised fine-tuning, even on physics-degraded data produced by a flight-calibrated simulator, applies no explicit pressure toward a sensor-invariant representation, and none emerges as a side effect. An MMD-minimization loss applied directly to the encoder during this fine-tuning stage was also explored as a more explicit alternative and is not pursued further in this thesis, superseded by the mechanism below, which exploits a structural asset – co-registration – that a generic MMD objective does not use.

### 3.4.2 Adopted mechanism: paired-contrastive triplet alignment

The co-registered triplet dataset provides something most domain-adaptation settings lack: for every training example, the identical underlying scene observed by two or three different sensors. This licenses a directly supervised alignment objective rather than an unpaired one: a contrastive loss that pulls the embeddings of matched real- $\Phi$ -sat-2, real-Sentinel-2, and simulated- $\Phi$ -sat-2 patches of the same scene together, while pushing apart embeddings of unrelated scenes, regardless of which sensor produced each embedding. This directly penalizes the encoder for representing sensor identity, rather than only scene content, and does so using the strongest available supervisory signal – exact scene correspondence – instead of matching marginal distributions between unpaired sensor domains as an MMD-style objective would.

## 3.5 Cross-Architecture Knowledge Distillation

TerraMind’s ViT-based encoder cannot run on the  $\Phi$ -sat-2 on-board processor: the Myriad 2 VPU’s supported operator set has no dynamic-axes support, which ViT inference requires regardless of model size. Compression must therefore cross architecture families, from ViT teacher to CNN student, rather than compress within the ViT family as some prior on-board GeoFM work does.

Two placements for the domain-invariance objective were considered. The first prebakes invariance into the TerraMind teacher via the contrastive mechanism above, then performs standard distillation, so the student inherits invariance passively by mimicking an already-invariant teacher. The second applies the paired-contrastive triplet loss

directly to the student’s own embeddings during distillation, so the student is trained for cross-sensor consistency directly, rather than only for matching the teacher’s outputs. This thesis adopts the second: the contrastive triplet loss is applied to the student CNN during distillation, alongside the standard distillation objective against the teacher’s outputs. This is the central methodological contribution of the thesis, and its evaluation does not depend on any prior baking-in of invariance at the teacher stage – the two placements make different, separable claims, and this thesis tests the stronger one, that distillation itself can be the mechanism that instills domain-invariance in the deployable model rather than a step that merely hopes to preserve it.

## 3.6 Evaluation Protocol

Three evaluations follow from the mechanisms above. First, LULC classification on real  $\Phi$ -sat-2 data, comparing the contrastive-KD-distilled student against a Sentinel-2-only baseline and against a student distilled with the standard response/logit-only objective alone, since real ground truth exists only for this task. Second, flood segmentation across the simulated degradation ladder, to characterize how gracefully performance degrades with increasing sensor-noise severity. Third, a few-shot label-efficiency comparison against  $\Phi$ satNet (Section 2.5.1) on matched small label budgets, testing directly whether adapting TerraMind’s broad, diverse pre-training generalizes to  $\Phi$ -sat-2 with fewer labels than  $\Phi$ satNet’s mission-specific pre-training requires – the empirical test of this thesis’s central bet, reported alongside the possibility that it does not hold (Section ??).

## 3.7 Tools, Software Environment, and Reproducibility

The degradation pipeline and dataset construction are implemented as modular `EOTask` objects within the `eo-learn` framework, orchestrated through a versioned `SimulationConfig` configuration object rather than hard-coded parameters, so that a given degradation run (baseline-calibrated or extended-sweep) is fully specified by its configuration rather than by code edits. Triplet and LULC datasets are stored in HDF5 with an accompanying manifest recording per-patch provenance and train/validation/test split assignment (fixed random seed, positional split), enabling exact reproduction of any reported split. Model training (TerraMind fine-tuning, contrastive-KD distillation) is implemented in PyTorch. The codebase is organized as a single versioned Python package (`terra_sat_drift`) with submodules separating data simulation, dataset loading, model training tasks, and drift analysis, under standard git version control, so that each experiment reported in Chapter 4 corresponds to a specific, retrievable commit and configuration rather than an undocumented one-off script.

# Chapter 4

## Results and Discussion

### 4.1 Practical Implementation

Two TerraMind variants, Tiny and Base, were fully fine-tuned on the real- $\Phi$ -sat-2 branch of the triplet dataset (Section 3.1.1) for the LULC task, following the per-domain normalization protocol of Section 3.3.1: square-root transform, smooth clip to a domain-specific value, and  $z$ -score standardization using statistics computed separately for each sensor domain. An initial evaluation pass understated performance because of a normalization defect causing weight saturation; all numbers reported below are post-correction. The simulated degraded Sen1Floods ladder (Section 3.1.2) was built at the flight-calibrated baseline degradation level and at an extended, beyond-nominal sweep, using the pipeline of Section 3.2, and TerraMind was fine-tuned and evaluated on each rung of that ladder independently. Domain-gap diagnostics (MMD, t-SNE, per-layer cosine similarity) were computed on frozen TerraMind embeddings extracted from the co-registered triplets, under both shared and per-domain normalization, to rule out normalization artifacts as an explanation for the measured gap (Section 3.3.1).

### 4.2 Domain Gap in Latent Space (RQ1)

#### 4.2.1 Real-data LULC baseline

TerraMind fine-tuned directly on real  $\Phi$ -sat-2 LULC data achieves mIoU 0.47 (Tiny) and 0.51 (Base) on the held-out real test set – a solid result given the scarcity of real, labeled  $\Phi$ -sat-2 data, and one obtained without any explicit domain-adaptation mechanism, since the model here is trained and evaluated on the same (real) sensor domain. This result establishes that the underlying task is learnable at this label budget and is not itself the bottleneck; the domain-gap question only arises when this same encoder is asked to transfer from Sentinel-2 alone.

**[TBD – experiment pending]** A Sentinel-2-only fine-tuned baseline, evaluated zero-shot on real  $\Phi$ -sat-2 LULC data, is required to complete the source-vs-target comparison in the framing used by related on-orbit GeoFM studies (Section 2), and to quantify the “before adaptation” gap that the contrastive-KD mechanism (Section 4.3) is evaluated against.

### 4.2.2 Where the gap concentrates

Domain-gap diagnostics on the co-registered triplets show that the gap is not uniform across TerraMind’s depth: cosine similarity between matched real- $\Phi$ -sat-2 and Sentinel-2 embeddings is lowest in the encoder’s early layers, and per-domain normalization sharpens rather than dissolves this early-layer separation relative to a shared-normalization baseline. Since a normalization artifact would be expected to shrink, not sharpen, apparent similarity once each domain is properly calibrated to its own statistics, this result rules out calibration mismatch as the source of the measured gap and indicates a genuine representational effect concentrated in the earliest stages of the encoder, plausibly reflecting low-level spectral and textural statistics that later, more semantic layers partially normalize away.

Histogram comparison of the two sensor domains points to differing spectral response functions between  $\Phi$ -sat-2 and Sentinel-2, rather than the spatial-resolution mismatch emphasized in related on-orbit GeoFM work (Section 2), as the likely dominant driver – notable because  $\Phi$ -sat-2’s ground sampling distance (4.75m) is finer than Sentinel-2’s (10–20m), the reverse of the coarser-target case that motivates a resolution-centric account of domain gap elsewhere.

**[TBD – experiment pending]** A controlled ablation isolating the individual contribution of spectral-response mismatch, band misalignment, signal-to-noise ratio, and optical blur to the measured gap has not yet been run; the spectral-response hypothesis above is drawn from histogram inspection, not from an ablation, and should be read as a working hypothesis pending that experiment.

## 4.3 Domain Adaptation and Distillation (RQ2, RQ3)

### 4.3.1 A negative result that reshaped the approach

Cross-validating against the real triplets shows that TerraMind fine-tuned on the physics-degraded Sen1Floods ladder, using standard supervised fine-tuning alone, does not reduce the domain gap to real  $\Phi$ -sat-2 relative to a model fine-tuned only on clean Sentinel-2 – even at the flight-calibrated degradation level, and even acknowledging that this comparison is partly confounded by task mismatch (Sen1Floods vs. the LULC task the triplets are labeled for). Standard fine-tuning on degraded examples applies no explicit pressure toward a sensor-invariant representation, and none emerges incidentally. This result, rather than a setback, is what motivated the paired-contrastive mechanism of Section 3.4.2, applied directly to the student during cross-architecture distillation rather than as encoder-level fine-tuning.

**[TBD – experiment pending]** Results for the adopted mechanism itself – the contrastive-KD-distilled CNN student’s LULC performance against the Sentinel-2-only baseline and against a student distilled with the standard response/logit-only objective alone – are the headline result this thesis is structured around (Section 4.1), and are pending completion of the corresponding training runs. The logit-only distillation baseline has been run and

judged qualitatively insufficient, consistent with the negative fine-tuning result above, but its numeric comparison against the contrastive-KD result is not yet available.

### 4.3.2 Robustness across the degradation ladder

Flood-segmentation IoU on the simulated Sen1Floods ladder degrades only mildly across the full range of simulated degradation severity, from 0.71 on the clean scene to 0.67 at the highest degradation level tested (including the lowest simulated signal-to-noise condition). This is a smaller effect than the LULC domain-gap results above might suggest, and is discussed further in Section 4.4.1.

## 4.4 Discussion

### 4.4.1 Interpreting the flood-robustness result

The mild flood-IoU degradation reported above is not unique to this thesis: Du et al. [11] independently report that flood detection was anomalously robust to the sensor domain gap on Kanyini, in contrast to a substantial (approximately 47%) drop in cloud-detection performance on the same platform. Two independent studies observing the same pattern is worth foregrounding rather than treating as incidental: water’s spectral signature (strong near-infrared absorption contrast) plausibly survives sensor noise and band mismatch better than the texture- and boundary-dependent cues that cloud or general land-cover classification rely on. This has a direct consequence for how this thesis’s motivating narrative should be read: flood segmentation, the downstream task chosen here, may understate the severity of the domain-gap problem this thesis’s method addresses, since it happens to be an unusually forgiving task. This is stated here as an acknowledged limitation of the choice of downstream task, not concealed as a favorable result.

### 4.4.2 Positioning against $\Phi$ satNet

$\Phi$ satNet (Section 2.5.1), developed within the same research group, reports that a purpose-built, non-transformer GeoFM trained from scratch on a mission-specific corpus matches or exceeds unconstrained transformer-based GeoFMs on several on-board tasks. Taken at face value, this raises a reasonable objection to the approach in this thesis: if a bespoke lightweight model already performs well without any of the domain-adaptation or distillation machinery developed here, what is gained by adapting TerraMind instead? This thesis’s answer is a label-efficiency bet, not a raw-performance claim: TerraMind’s large-scale, diverse, multi-modal pre-training is expected to generalize to new  $\Phi$ -sat-2 downstream tasks with fewer labels than  $\Phi$ satNet’s narrower, curated pre-training task suite requires. The intended test of this bet – a matched few-shot comparison against  $\Phi$ satNet on a shared task and label budget – is not yet run.

**[TBD – experiment pending]** Few-shot comparison against  $\Phi$ satNet (or its published OnboardEO-Bench numbers, where a matched task and label budget exists) is planned but not yet executed.

Critically, this thesis’s central methodological claim does not depend on the outcome of that comparison. The contribution under test is the paired-contrastive, co-registration-exploiting distillation mechanism itself (Section 3.4.2) – a general technique for closing sensor domain gap during cross-architecture compression – which remains valid and useful regardless of whether TerraMind ultimately proves more or less label-efficient than a bespoke model such as  $\Phi$ satNet on this specific mission. The backbone-choice question and the method-validity question are deliberately kept separable.

### 4.4.3 Limitations

This work is scoped to a single mission and sensor pair ( $\Phi$ -sat-2 and Sentinel-2); no comparison to other on-board missions or sensor combinations is attempted, so the generality of the specific spectral-response-driven gap finding (Section 4.2) beyond this pair is untested. The thesis also relies on TerraMind as a given, externally pre-trained GeoFM rather than training a custom foundation model, which is a deliberate scope decision (Section 1.3) but means the label-efficiency bet in Section 4.4.1 is tested for one specific large GeoFM, not for the class of large GeoFMs in general. The flood-segmentation downstream task, as discussed above, may understate rather than represent the severity of the domain-gap problem for other applications.

### 4.4.4 Practical recommendations

Two recommendations follow directly from this work. First, missions intended to host on-board GeoFMs should prioritize collecting co-registered calibration data – across the sensor pair the mission actually needs to bridge – early in the mission’s operational life, rather than waiting for labeled real-world event data to accumulate; this is precisely what made the domain-gap measurement in this thesis possible at all. Second, physics-based degradation simulation should be treated as a necessary complement to real paired data, not a substitute for it: the negative fine-tuning result (Section 4.3) shows that simulated data alone, however well calibrated, does not automatically confer robustness. Concrete directions for narrowing the remaining simulation-to-real gap include histogram matching of simulated output against real target-sensor statistics, and calibrating the simulator against known ground-truth structures visible in both sensors.

# Chapter 5

## Conclusion

On-board processing exists to shorten the path from acquisition to actionable information for time-critical Earth observation applications such as flood and wildfire response, where the value of a detection decays with every hour spent waiting for a ground-based product. Because a mission’s own sensor data cannot exist before launch, and labeled examples of rare events accumulate slowly afterward, foundation models pre-trained on other missions’ archives are an attractive way to reach useful performance with few mission-specific labels. This attractiveness is conditional, however, on a question this thesis set out to answer directly: whether such a model’s learned representation actually survives contact with a sensor it was never trained on, and, if not, whether that can be fixed deliberately rather than assumed away.

The evidence gathered here says it can, but not for free. TerraMind’s latent representation of real  $\Phi$ -sat-2 imagery does differ measurably from its representation of Sentinel-2 imagery of the same scenes, concentrated in the encoder’s early layers and driven primarily by differing spectral response between the two sensors rather than by their difference in spatial resolution. Naively fine-tuning on physics-simulated degraded data does not close this gap; an explicit training signal that exploits co-registered sensor pairs directly is required, and this thesis applies that signal during the same step – knowledge distillation – that is separately required to compress the model onto space-qualified hardware that cannot run its original Vision Transformer architecture at all. Treating the domain-gap problem and the compute-constraint problem as one combined training objective, rather than as two sequential and independent fixes, is the thesis’s central methodological position.

This work is limited to a single mission and sensor pair, and to one specific pre-trained GeoFM rather than a family of them; whether the spectral-response-driven account of the domain gap generalizes to other on-board missions, and whether TerraMind’s broad pre-training is in fact more label-efficient than a mission-specific model trained from scratch, are both open questions this thesis frames precisely but does not close. The comparison against  $\Phi$ satNet, a bespoke GeoFM developed for the same mission by the same research group, is the immediate next step toward closing the second of these, and a natural extension of this work would repeat the analysis for other sensor pairs and other missions, or attempt to train a custom foundation model rather than adapt an existing one, to test whether the findings here are specific to TerraMind or general to the class of large, diversely pre-trained EO models it represents.

# Bibliography

- [1] D. Barretta *et al.*, “Towards real-time Earth system intelligence through advancements in onboard AI processing,” *TechRxiv preprint*, 2026. Preprint, not yet peer-reviewed as of drafting – [VERIFY current publication status before submission; prioritize the peer-reviewed version if one appears.].
- [2] B. Tellman, J. A. Sullivan, C. Kuhn, A. J. Kettner, C. S. Doyle, G. R. Brakenridge, T. A. Erickson, and D. A. Slayback, “Satellite imaging reveals increased proportion of population exposed to floods,” *Nature*, vol. 596, no. 7870, pp. 80–86, 2021.
- [3] S. W. Taylor, D. G. Woolford, C. Dean, and D. L. Martell, “Wildfire prediction to inform fire management: statistical science challenges,” 2013.
- [4] Intel Movidius, “Myriad 2 vision processing unit – product brief,” tech. rep., Intel Corporation, 2016. [VERIFY: cite the specific Intel/Movidius datasheet or product brief you actually used, with its own date/URL access note per your university’s citation guidelines on non-journal sources.].
- [5] C. Coelho *et al.*, “NanoSat MO Framework: Enabling AI apps for Earth observation,” in *35th Annual Small Satellite Conference*, no. SSC21-S1-31, 2021. [VERIFY full author list against the conference proceedings.].
- [6] A. Marin, C. Coelho, F. Deconinck, I. Babkina, N. Longepe, and M. Pastena, “Phisat-2: Onboard ai apps for earth observation,” *Proc. Space Artif. Intell.*, vol. 6, 2021.
- [7] J. Jakubik *et al.*, “TerraMind: Large-scale generative multimodality for Earth observation,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025. arXiv:2504.11171. [VERIFY full author list against the ICCV/arXiv record.].
- [8] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, *et al.*, “An image is worth 16x16 words: Transformers for image recognition at scale,” *arXiv preprint arXiv:2010.11929*, 2020.
- [9] N. Longép   *et al.*, “Simulation of multispectral and hyperspectral EO products for onboard machine learning application,” *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 17, pp. 17651–17665, 2024. [VERIFY full author list against the IEEE Xplore record.].
- [10] G. Hinton, O. Vinyals, and J. Dean, “Distilling the knowledge in a neural network,” *arXiv preprint arXiv:1503.02531*, 2015. Presented at the NIPS 2014 Deep Learning Workshop.

- [11] B. Du *et al.*, “First on-orbit demonstration of a geospatial foundation model,” *arXiv preprint arXiv:2512.01181*, 2025. [VERIFY full author list against the arXiv record.].
- [12] W. M. Kouw and M. Loog, “An introduction to domain adaptation and transfer learning,” *arXiv preprint arXiv:1812.11806*, 2019.
- [13] A. Du, A.-D. Doan, Y. W. Law, and T.-J. Chin, “Domain adaptation for satellite-borne multispectral cloud detection,” *Remote Sensing*, vol. 16, no. 18, p. 3469, 2024.
- [14] G. Giuffrida, L. Fanucci, G. Meoni, M. Batič, L. Buckley, A. Dunne, C. Van Dijk, M. Esposito, J. Hefele, N. Vercruyssen, *et al.*, “The  $\phi$ -sat-1 mission: The first on-board deep neural network demonstrator for satellite earth observation,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–14, 2021.
- [15] P. Lindenberger, P.-E. Sarlin, and M. Pollefeys, “LightGlue: Local feature matching at light speed,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.